WorkForce AI Pro

HR Attrition Prediction & Workforce Intelligence Platform

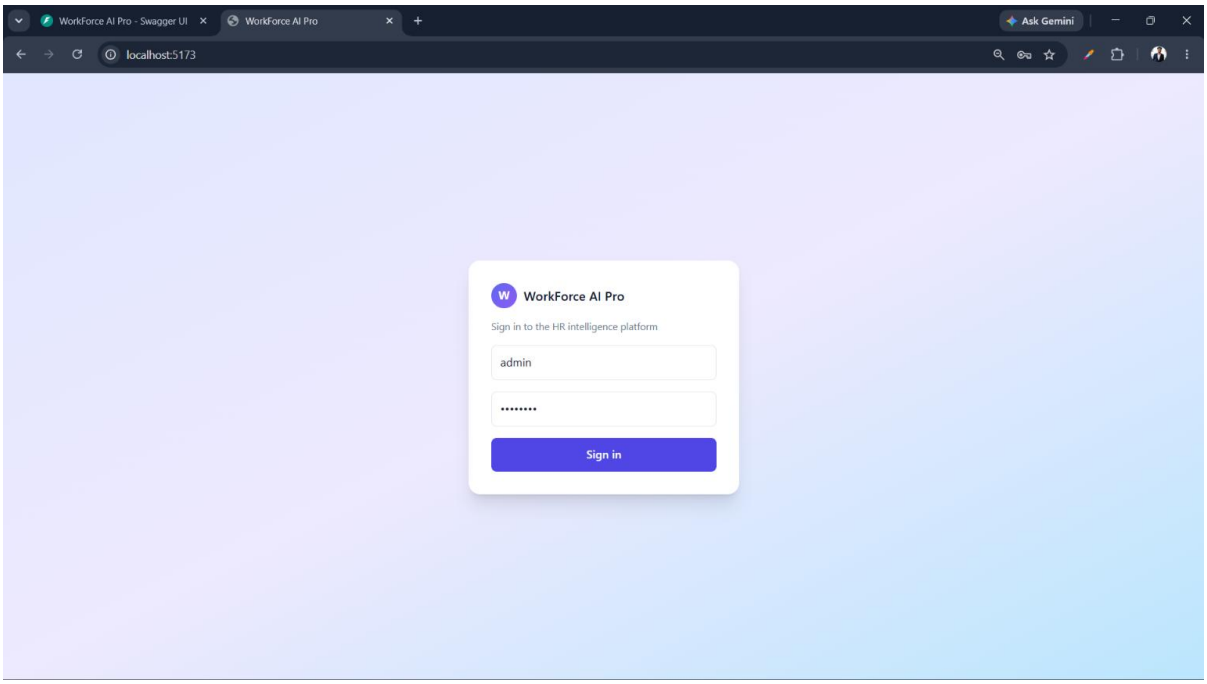
Submitted By: Dineshkumar S
Date: 27/10/2026

Requirement Completion

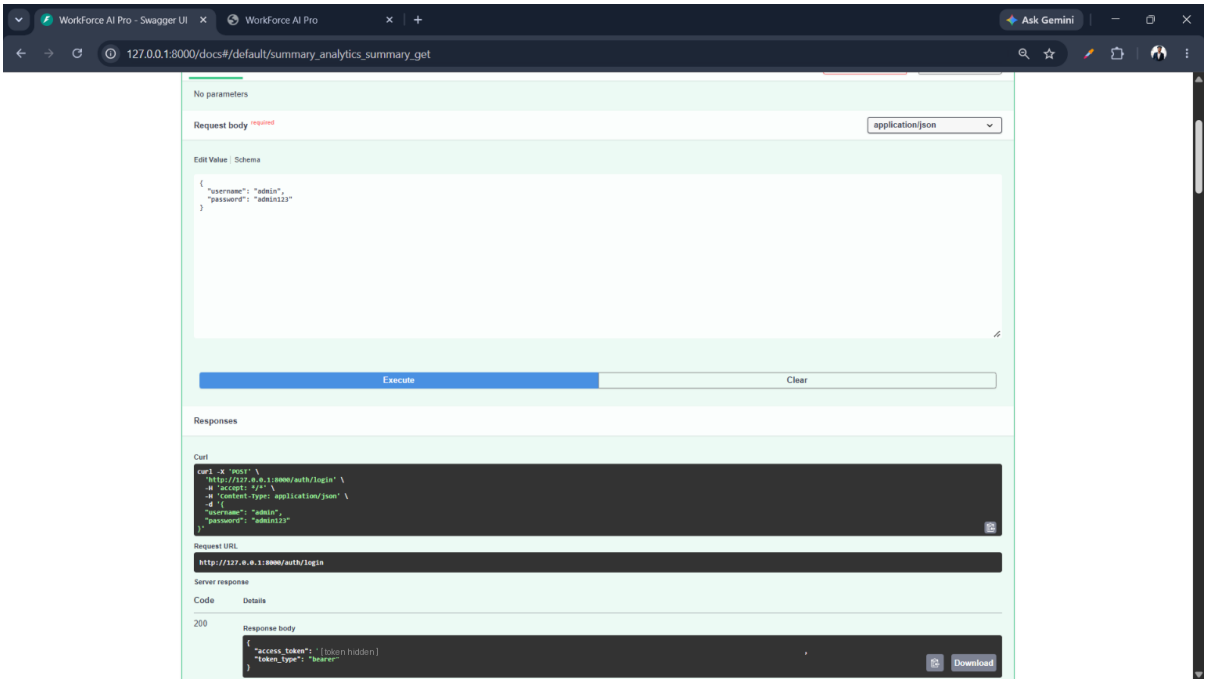
Requirement	Status	Evidence
Backend Source Code	Done	FastAPI backend with SQLAlchemy models, Pydantic schemas, JWT security and 11 API endpoints.
Frontend Source Code	Done	React (Vite) app with Tailwind CSS, Axios and Recharts: login, dashboard, employees, alerts, logs and risk predictor pages.
Database Schema	Done	MySQL database workforce_ai with 5 tables: users, employees, alerts, interventions and activity_logs.
JWT Authentication	Done	Login API returns an access token (200); a request without a token returns 401 Unauthorized.
AI Prediction APIs	Done	POST /predict returns a risk score and level (0.954, High for the sample employee); the Risk Predictor page shows the result.
Workforce Dashboards	Done	Dashboard with KPI cards, department breakdown, risk trends and recent alerts for 30 employees; employee risk register.
Alert and Intervention Workflow	Done	Employees with risk of 70% or more raise an HR alert and start an intervention workflow; alerts can be started or closed.
Monitoring Logs	Done	Every risk recalculation, alert and intervention is logged and listed with time and employee.
Analytics Reports	Done	Employee risk report exported as CSV with department, satisfaction, overtime, risk score and risk level.
API Documentation	Done	Swagger UI documents all 11 endpoints.

Evidence

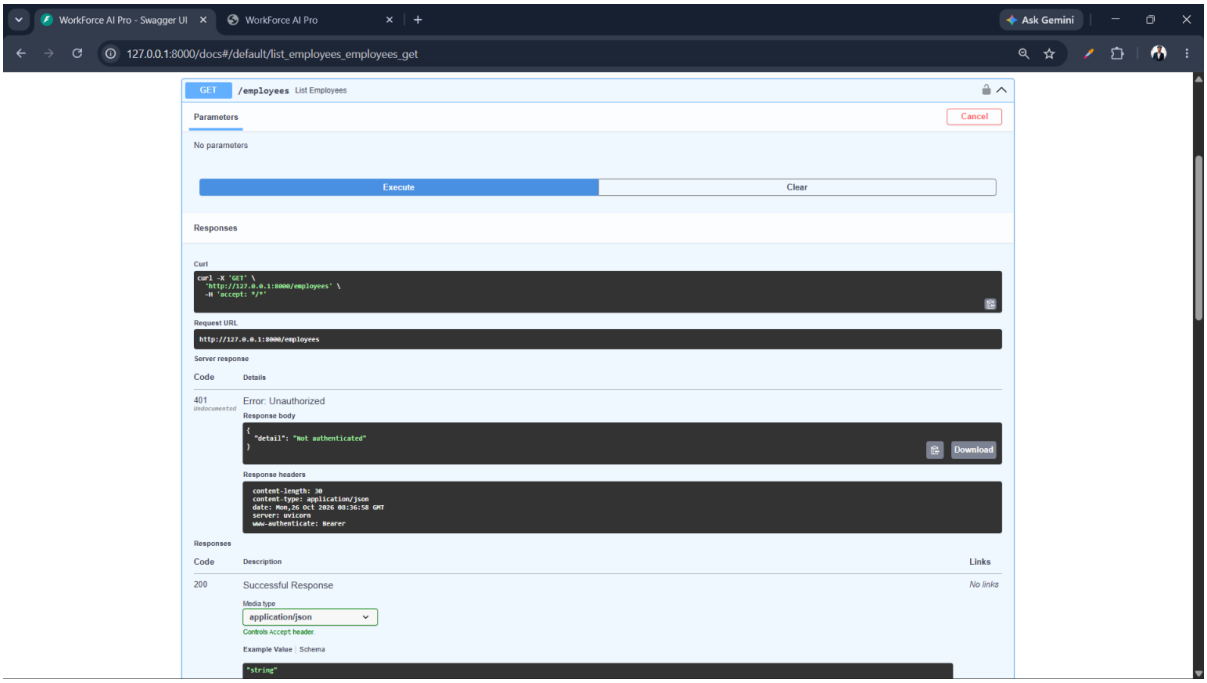
Authentication — Login Page



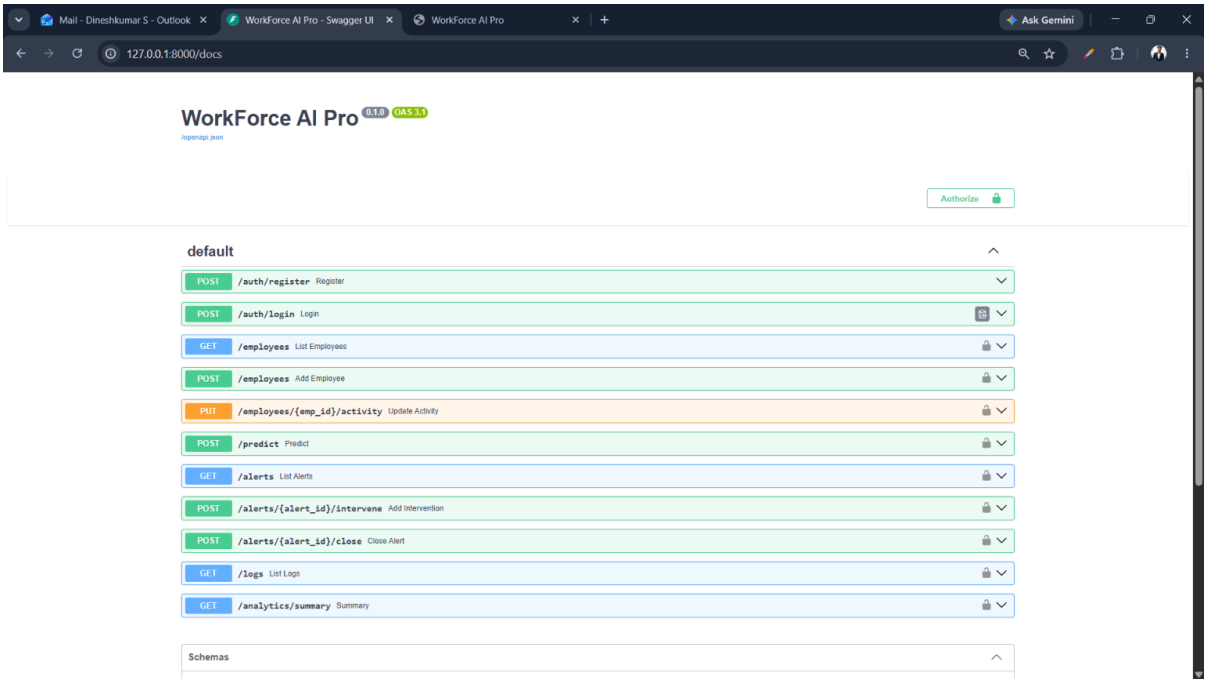
Authentication — Login API Returns JWT Access Token, 200 OK (token hidden)



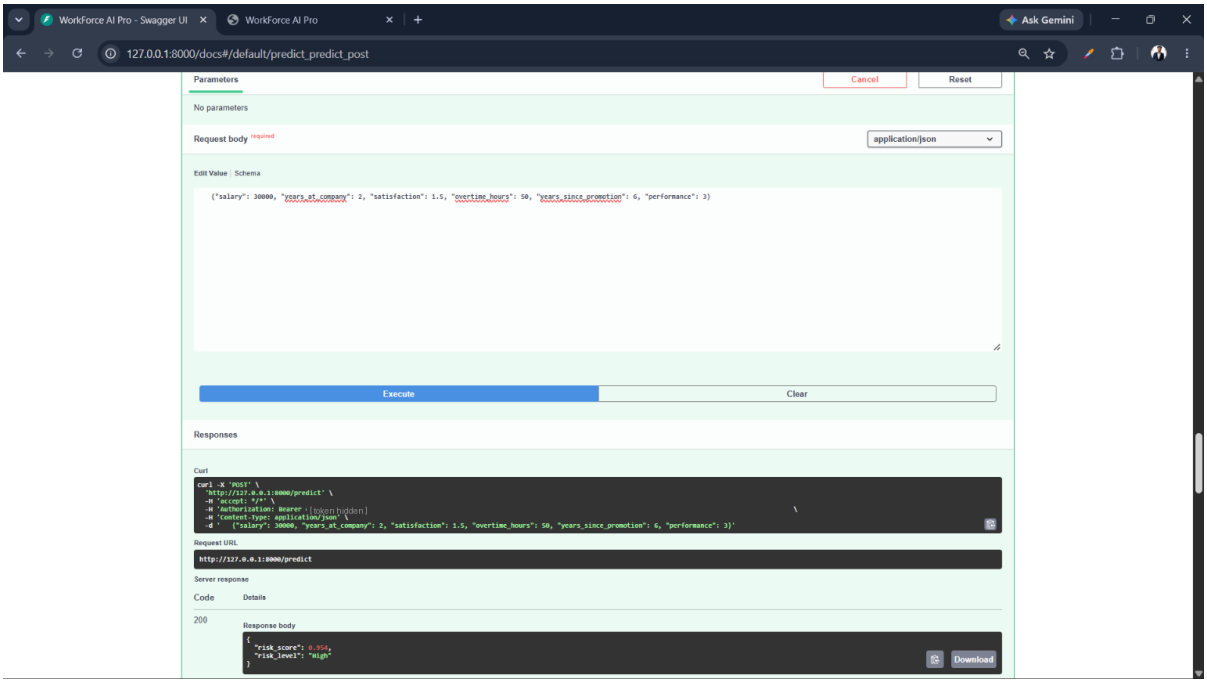
Security — Request Without Token Rejected, 401 Unauthorized



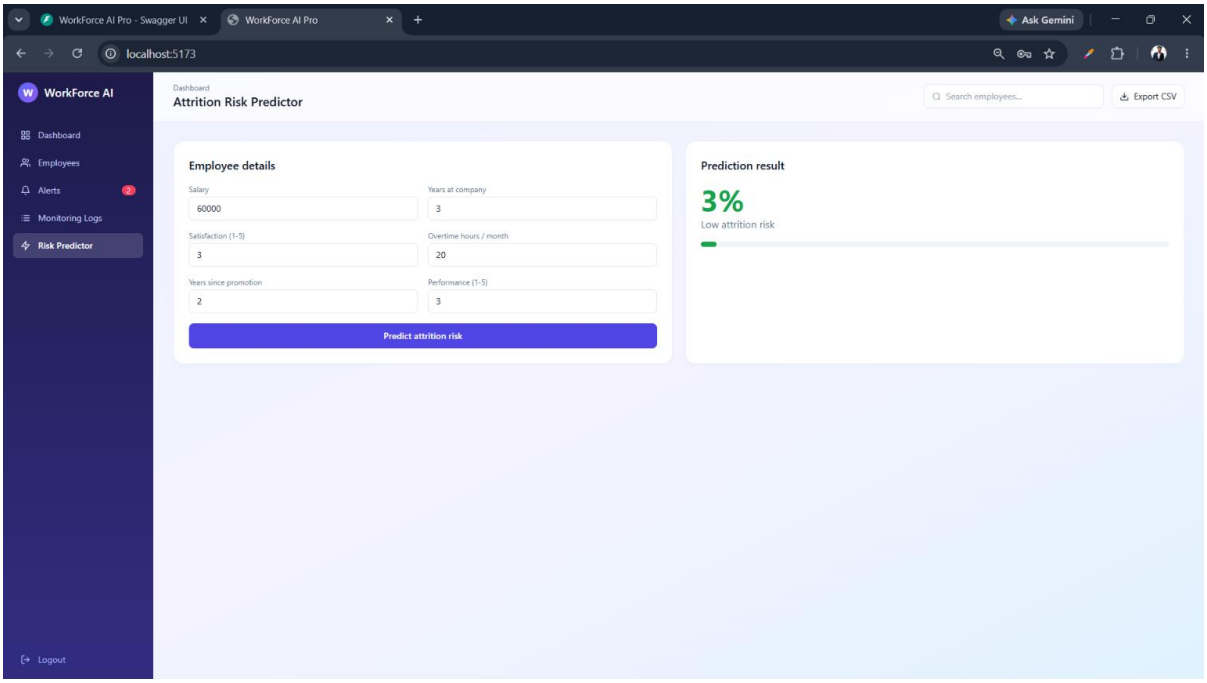
API Documentation — Swagger UI Endpoint List



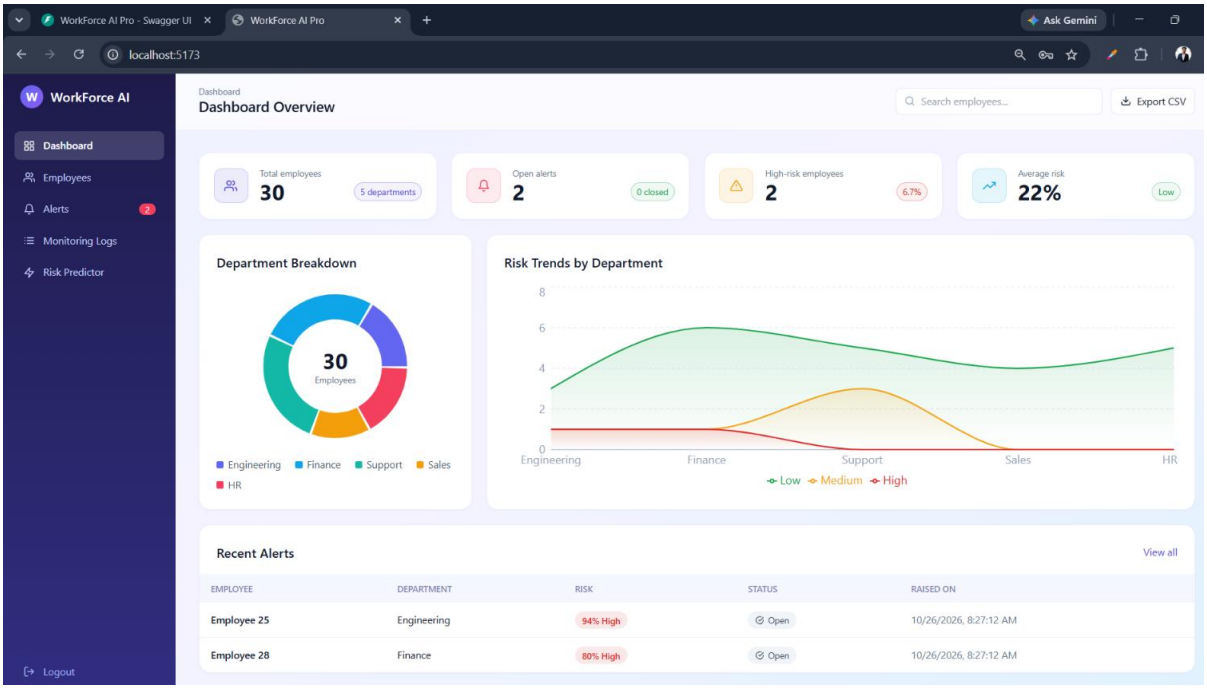
AI Prediction API — High-Risk Employee Scored 0.954, High (token hidden)



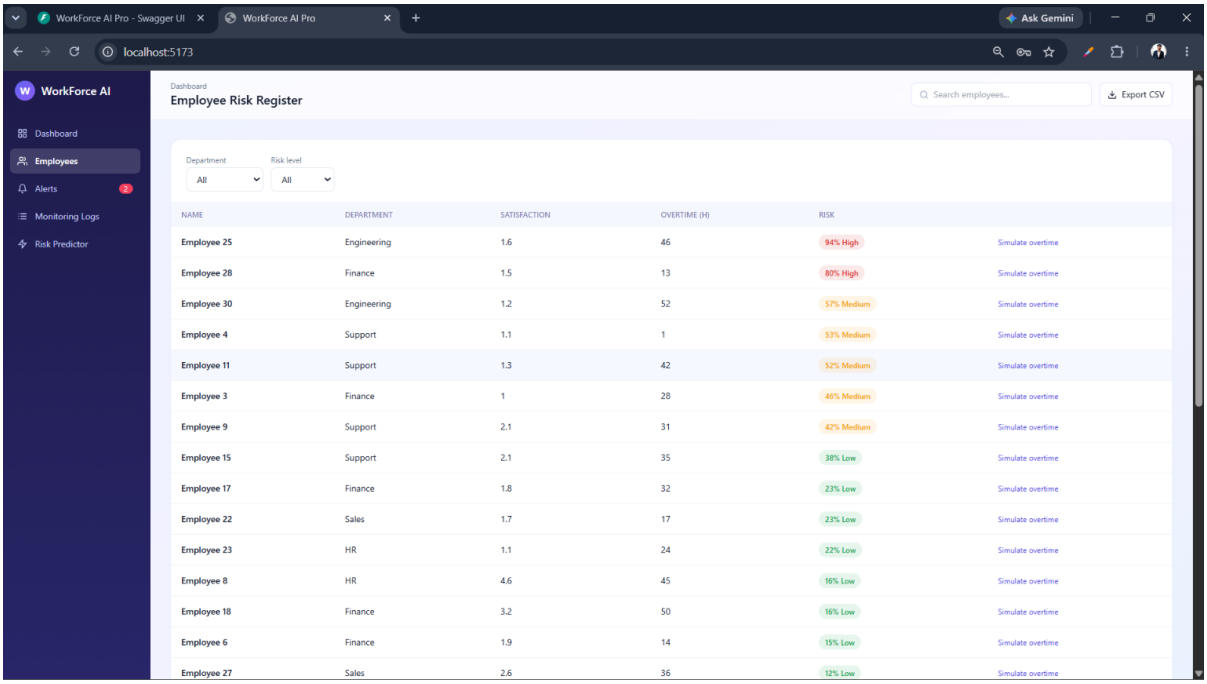
AI Prediction — Risk Predictor Page



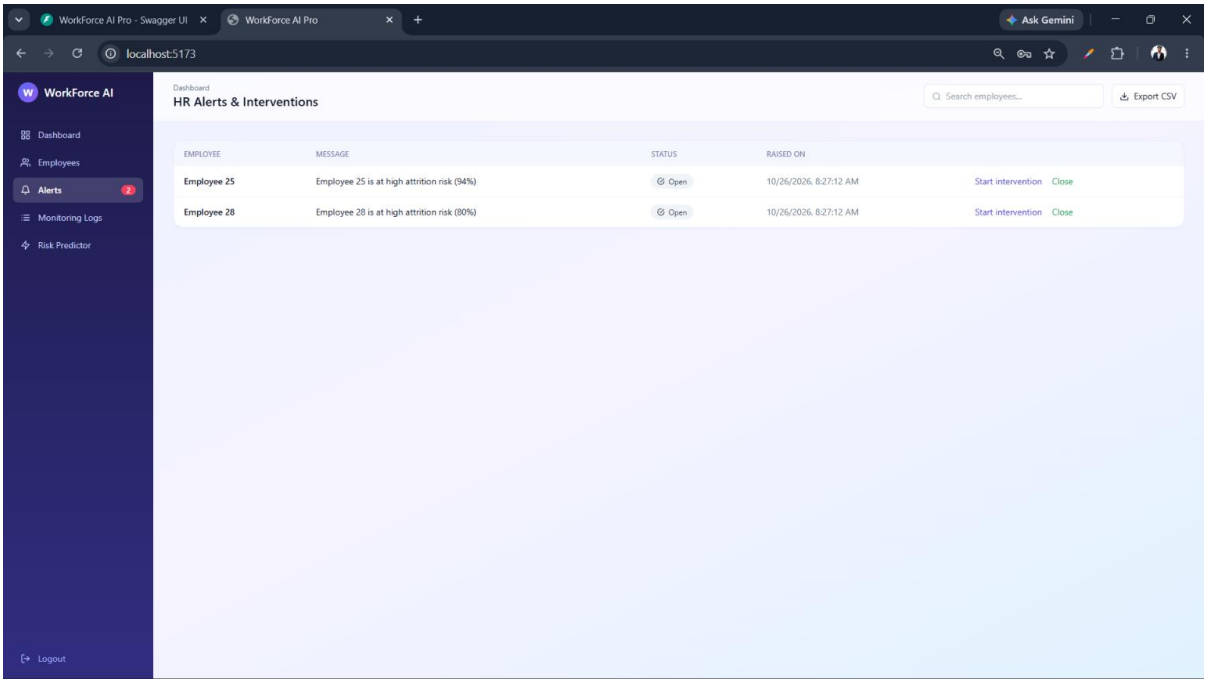
Dashboard — Organisation Overview (30 Employees)



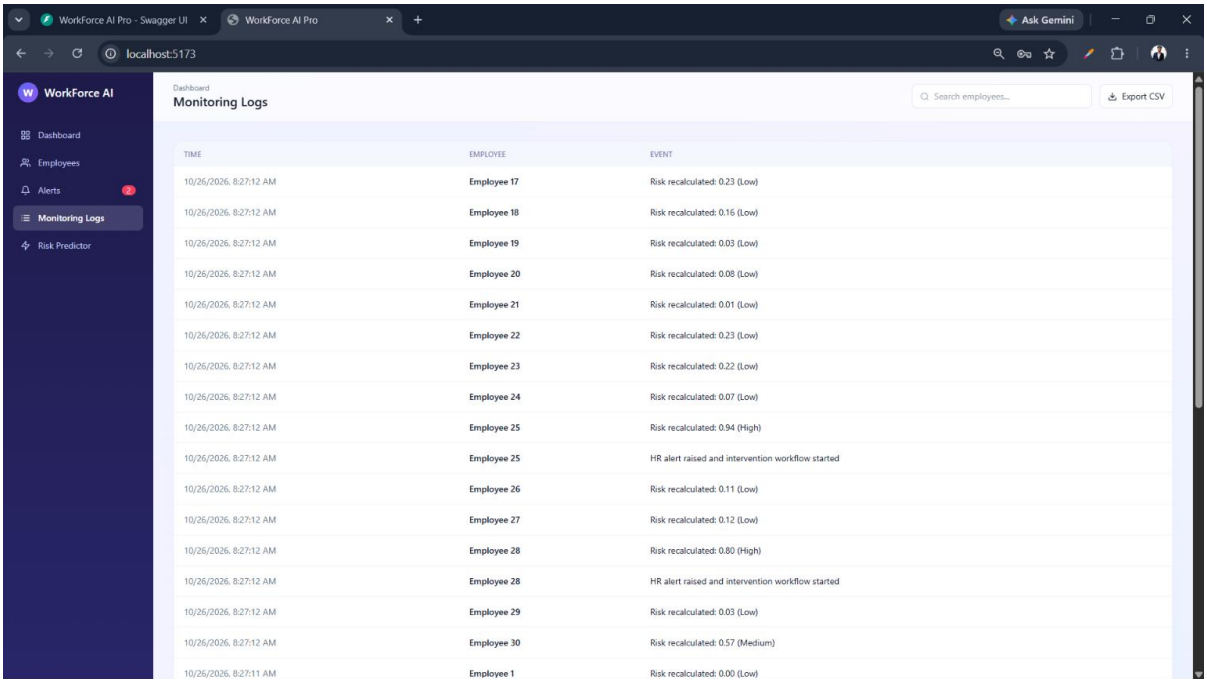
Employees — Risk Register Sorted by Predicted Risk



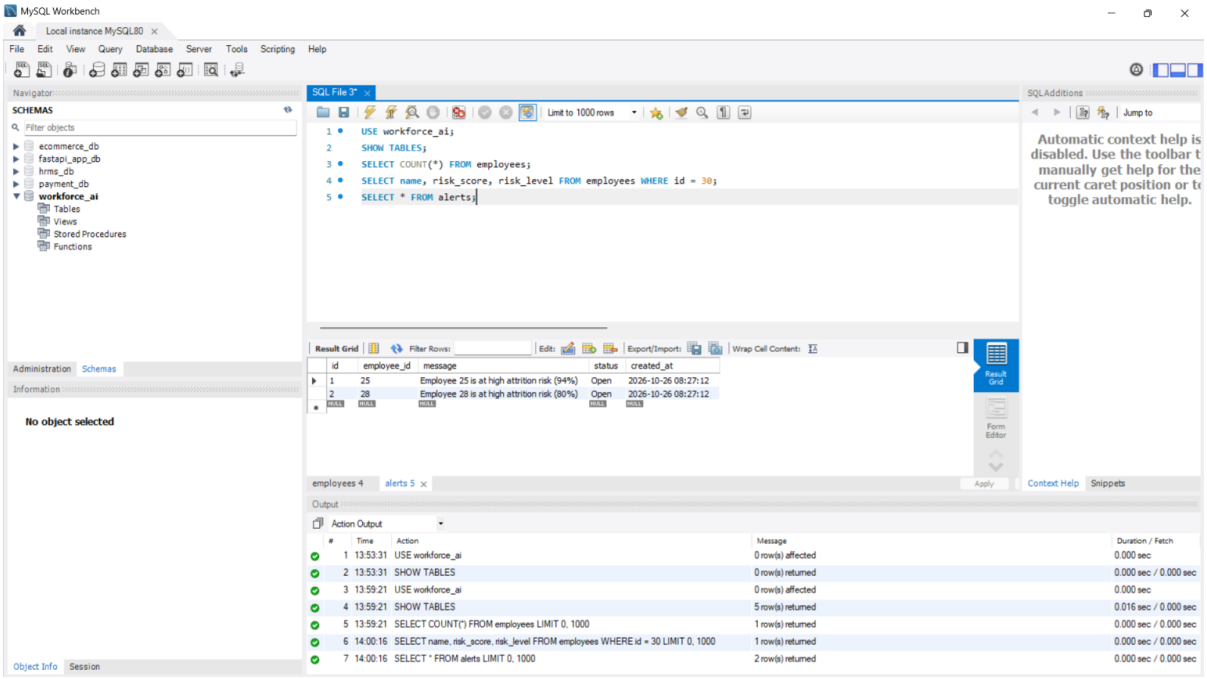
Alerts — HR Alerts Raised for High-Risk Employees



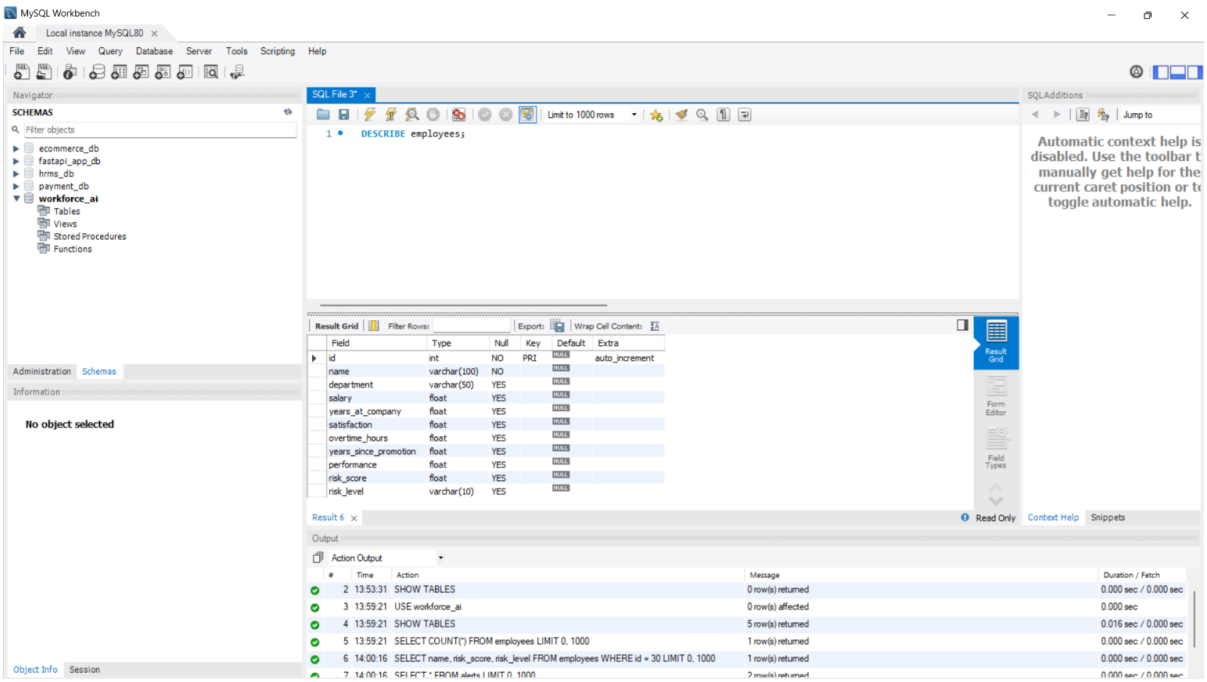
Monitoring — Logs of Risk Recalculation and Alert Workflow



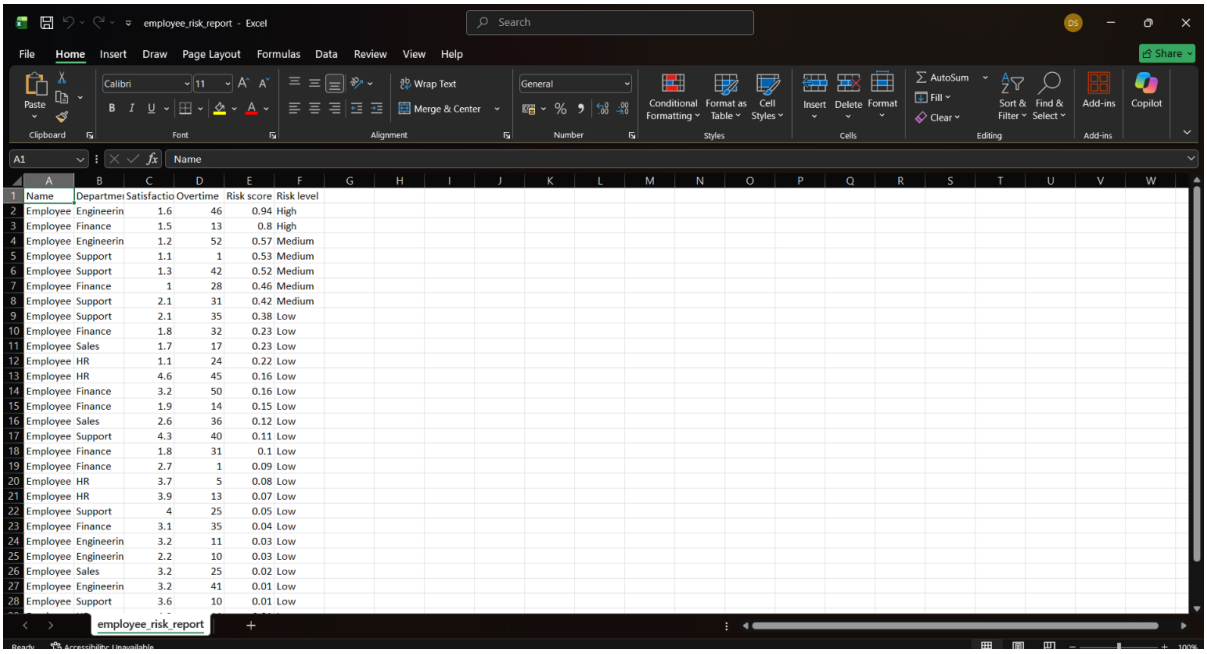
Database — MySQL alerts Table Storing HR Alerts



Database — employees Table Schema (MySQL)



Analytics — Employee Risk Report Exported as CSV



Name	Departmer	Satisfactio	Overtime	Risk score	Risk level
Employee	Engineerin	1.6	46	0.94	High
Employee	Finance	1.5	13	0.8	High
Employee	Engineerin	1.2	52	0.57	Medium
Employee	Support	1.1	1	0.53	Medium
Employee	Support	1.3	42	0.52	Medium
Employee	Finance	1	28	0.46	Medium
Employee	Support	2.1	31	0.42	Medium
Employee	Support	2.1	35	0.38	Low
Employee	Finance	1.8	32	0.23	Low
Employee	Sales	1.7	17	0.23	Low
Employee	HR	1.1	24	0.22	Low
Employee	HR	4.6	45	0.16	Low
Employee	Finance	3.2	50	0.16	Low
Employee	Finance	1.9	14	0.15	Low
Employee	Sales	2.6	36	0.12	Low
Employee	Support	4.3	40	0.11	Low
Employee	Finance	1.8	31	0.1	Low
Employee	Finance	2.7	1	0.09	Low
Employee	HR	3.7	5	0.08	Low
Employee	HR	3.9	13	0.07	Low
Employee	Support	4	25	0.05	Low
Employee	Finance	3.1	35	0.04	Low
Employee	Engineerin	3.2	11	0.03	Low
Employee	Engineerin	2.2	10	0.03	Low
Employee	Sales	3.2	25	0.02	Low
Employee	Engineerin	3.2	41	0.01	Low
Employee	Support	3.6	10	0.01	Low